A FRIENDLY LEARNING GUIDE | MINING RAIL OPERATIONS + OPTIMIZATION

Rail Crew Recovery in Mining

From a late train to your first MILP model,
explained in a simple, conversational way.

Operations | Real field problems | Feasibility | Assignment | MILP

Teaching examples only. Not confirmed company rules, data or an internal model. Always check real operator rules.

Welcome

WHAT YOU WILL LEARN	HOW TO USE THIS GUIDE
What crew scheduling and crew recovery are, what goes wrong in a real mining railway, how to check if a crew can take a train (time, place, rest, buffer), and how to turn that into a small optimization (MILP) model.	Read it like a chat. When the Coach asks a question, **pause and answer first**, then read the answer. The "Say it like this" boxes help you explain ideas clearly in meetings and interviews.

Scope note

Every example here is a simplified learning example. These are **not** confirmed company rules, data or an internal model. Real duty, rest, sign-on and sign-off rules must always be checked in the operator's own documents. Items marked **Field reality** describe common situations in heavy-haul and mining railways in general.

Contents

1	The big picture: from mine to port
2	What really goes wrong in the field
3	The recovery cycle, step by step
4	Travel time, rest time and buffer time
5	Crew availability and crew location
6	Feasibility matrix and crew assignment
7	Turning it into a MILP
8	When a late train hits the next assignment
9	From toy model to real model
10	Key terms, quick answers and roadmap

The story in one minute

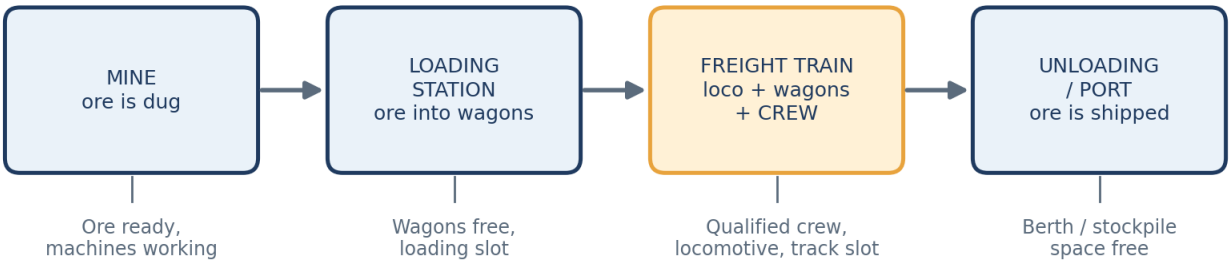
A mining railway moves ore from mine to port. Crews are scarce, trips are long and everything is linked. When something goes wrong (a crew is absent, a train is late), someone has to quickly find a new crew plan that is **safe, legal and cheap**. This guide teaches you how to think about that problem, step by step.

1 The big picture: from mine to port

Why a train needs more than a locomotive and wagons

COACH

Let's start simple. In mining, ore moves in a chain: **mine, loading station, train, unloading station or port**. Every step must be ready at the right time. And here is the part people forget: a train also needs an **available, qualified crew**. No crew means no movement.



The crew must be in the right place, at the right time, with the right rest.

Figure 1. A simplified mining freight chain. A train movement needs ore, equipment, track, and a ready crew.

COACH

Two words you will hear all the time: **crew scheduling** is the advance plan (which crew runs which train, and when). **Crew recovery** is what you do when that plan breaks, for example when a crew becomes unavailable or a train arrives late.

Think of it like a school bus

Scheduling is the timetable printed at the start of the term. Recovery is what the transport manager does at 6:30 AM when a driver calls in sick and the bus is already late. Same people, same buses, but now you must decide fast.

Knowledge check

Try all five before you look at the answers below.

COACH

Q1. Why can't a train run just because the locomotive and wagons are available?
Q2. What is the difference between crew scheduling and crew recovery?
Q3. C1 is assigned to T1 but becomes unavailable. What is the immediate problem?
Q4. Name three things we may want to minimize during recovery.
Q5. If there are five trains but only three crews, what challenge might we face?

YOUR ANSWER

1. Because train operate when goods are available and it's has to transport.
2. Crew scheduling means telling which crew is going to assign which train at what time but crew recovery is finding best crew schedule if some crew are not available.
3. Crew recovery.
4. Minimize train delay, minimize crew overtime, and minimize cost.
5. Train delay will increase. We may have to wait for a crew to finish and then relocate them from another train, or possibly outsource.

SAY IT LIKE THIS

A train can operate only when the goods **and** a qualified crew are available.
Crew scheduling means deciding which crew will be assigned to each train and at what time. Crew recovery means finding a revised crew schedule when some crews become unavailable or disruptions occur.
The immediate problem is that T1 may no longer have an available crew, so it may be delayed. Crew recovery is the process used to find a solution.
We should minimize train delays, crew overtime, and operating costs.
Train delays may increase. We may need to wait until a crew finishes its current duty and then reposition that crew, or consider another permitted resource.

COACH SAYS

Good start. The key idea is to separate the **operational problem** (a train has no available crew) from the **recovery process** (finding a revised plan). For Q3 the answer is the problem itself: T1 is now uncovered. Crew recovery is the *solution process*, not the problem.

2 What really goes wrong in the field

The disruptions that make crew recovery necessary, and why mining makes them harder

COACH

Before any maths, let's look at the real world. Nobody builds a recovery model for fun. It exists because things go wrong every single day. Here are the common ones. Which of these would hurt you most, do you think? Keep the answer in mind as you read.

Field reality (general industry context)

The tables in this chapter describe typical situations in heavy-haul and mining railways. Names, limits and rules differ by country, operator and union agreement. Use them to build intuition, not as facts about any one company.

A. Problems on the crew side

What happens	Why it hurts	Typical example
Crew absent or late	The train has nobody to drive it	Sickness, transport delay to the sign-on point, a missed shift
Crew runs out of legal hours	A late train keeps the crew on duty longer than allowed, so they must stop	Train stuck 3 hours waiting for a free berth at the port
Not qualified for this train	Crew may only be approved for certain routes, locomotive types or train lengths	Only some crews are certified for a steep section or a heavy-haul loco
Crew in the wrong place	Available, but far from the departure station	Crew finished at the port end, next train starts at the mine end
Rest not completed	Rest rules stop a crew from starting again too soon	Crew arrived 4 hours ago but needs a longer rest after a long duty

B. Problems on the train and network side

What happens	Why it hurts	Typical example
Late loading at the mine	Train leaves late, arrives late, next crew plan shifts	Loader breakdown, ore not ready, long queue at the loading point

What happens	Why it hurts	Typical example
Locomotive or wagon fault	Train stops or is replaced; crew time is wasted	Loco failure in the middle of a long single line
Single track and crossing loops	Trains must wait for opposite trains to pass	One slow train blocks several others behind it
Track maintenance or incident	Section closed, trains held or rerouted	Planned possession overrun, derailment, level-crossing incident
Weather	Speed limits or stops add hours to a trip	Heavy rain, cyclone, extreme heat, bush fire near the track
Slow unloading at the port	Trains queue, crews wait, hours burn	Ship delayed, berth busy, stockpile full

C. Why mining makes it harder than a passenger railway

- **Very long trips.** One run can take many hours, so a small delay uses up a big part of the legal duty time.
- **Remote locations.** A replacement crew may be hours away by road, and there may be no easy way to move them.
- **Few crews, many trains.** There is often a thin pool of qualified drivers, so spare capacity is small.
- **Cycle operations.** The same train loops mine, port, mine, port. One late cycle delays the next cycle, and the one after.
- **Heavy business pressure.** Ships have time windows. Missing them can cost real money (often called demurrage), so some trains are far more important than others.
- **Shared resources.** Crew, locomotive, wagons and track slot are all linked. Fixing the crew alone can still leave the train stuck.

D. Your recovery options (the menu)

COACH When a crew problem appears, control staff usually pick from a menu. The model's job is to choose the **best** item from this menu quickly and safely.

Option	What it means	Watch out for
Swap with another crew	A different qualified crew covers the train	That crew may have been planned for another train
Use a standby / spare crew	A crew held in reserve takes over	Reserve crews are limited and may be far away
Deadhead a crew	Move a crew as passengers to the right station	Travel time and cost, but crew does not drive
Delay the train	Hold the train until a crew is ready	Delay cost, effect on later trains and ship windows
Extend duty (if allowed)	Ask the crew to work a bit longer	Only if rules explicitly allow it; fatigue and safety risk
Re-sequence trains	Change the order trains leave	One train gets earlier, another gets later
Cancel or merge a run	Drop a low-value movement	Lost tonnage, contract effects
Outsource	Use an external resource	Cost, availability, approvals

Hard rules vs soft goals

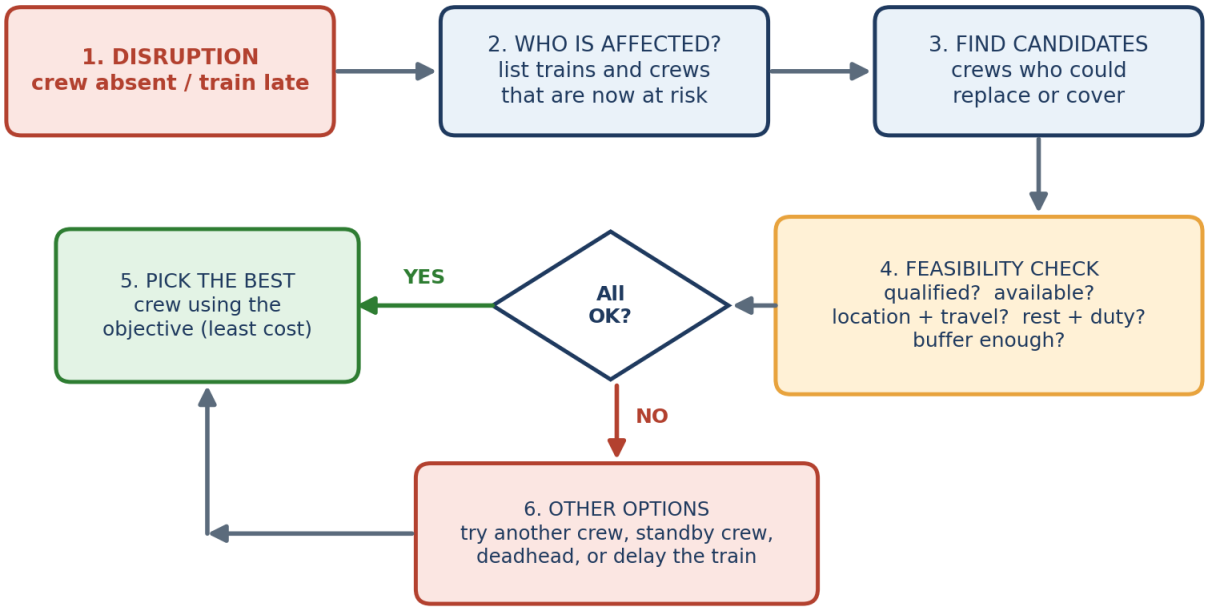
Hard constraints must never be broken (safety, legal rest, qualification). **Soft goals** are things we prefer (less delay, fewer changes, lower cost). The optimizer tries to get the best soft result while always obeying every hard rule.

3 The recovery cycle, step by step

What happens from the moment a disruption appears

COACH

Here is the whole loop in one picture. Don't worry about memorizing it. Just notice that it always runs the same way: something breaks, we find who can help, we test if they truly can, and we pick the best one.



Then: publish the new plan, tell crews and control, and watch for the next disruption.

Figure 2. A simplified decision flow. A real system may contain more checks, rules and recovery options.

Let's walk through it in plain words

Step	In plain words	Example
1. Disruption	Something changes the plan	C1 calls in sick; train T1 is stuck 45 min late
2. Who is affected	Find every train and crew touched by the change	T1 has no crew; C1's next train T4 is also at risk
3. Find candidates	List crews that might cover	C2, C3, and a standby crew at Station B
4. Feasibility check	Test each candidate against the rules	Qualified? Available? Can reach? Rested? Enough buffer?
5. Pick the best	Among feasible crews, choose the lowest cost or delay	C2 covers T1; no other train is hurt
6. Other options	If nobody is feasible, use the menu from Chapter 2	Delay T3 (low priority), keep T1 and T2 running

Do not skip step 4
A common beginner mistake is to pick the nearest crew. Nearest is not the same as feasible. A nearby crew may be unqualified, still resting, or about to run out of legal hours. Check first, then compare.

4 Travel time, rest time and buffer time

The three clocks every crew has to beat

COACH	Imagine C1 drives T1 from 8:00 AM to 11:00 AM. T2 leaves from a **different** station at 11:15 AM. Can C1 drive T2? Think about three things: travel time, required rest, and the chance that T1 is late.
YOUR ANSWER	In 15 min C1 can't reach to that station also he need some rest in between and may be train can be delayed so it's not possible for C1 to operate other train at 11:15.
SAY IT LIKE THIS	C1 cannot reach the other station in 15 minutes. The crew also needs some rest between duties, and T1 might be delayed. Therefore, C1 cannot operate the other train at 11:15 AM.
COACH SAYS	Correct. Check travel time, required rest and possible delay. This is a **feasibility check**: can the crew legally and practically take the next assignment?
COACH	Next case. C1 finishes T1 at 11:00 AM, needs **45 minutes** to reach T2's station and **30 minutes** of rest. T2 leaves at **1:00 PM**. Can C1 operate it?
YOUR ANSWER	Yes in that case it is possible to operate because 11:00 + 45 min + 30 = 12:15 so he has 45 min gap it is possible.
SAY IT LIKE THIS	Yes, in this case C1 can operate T2. The travel time and rest add up to 75 minutes, so C1 is ready at 12:15 PM. There is a 45-minute buffer before T2 departs at 1:00 PM.
COACH SAYS	Exactly. **Buffer time** is the extra time between the crew becoming ready and the next required reporting or departure time. If T1 is delayed by 30 minutes, the buffer drops from 45 to 15 minutes. Sign-on or preparation time may also need to be included.

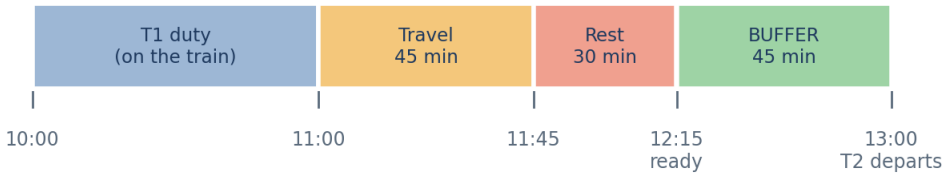


Figure 3. The 75 minutes of travel and rest leave a 45-minute safety buffer before T2 departs.

Time component	Illustrative value	What it means
T1 finishes	11:00 AM	C1 completes the first duty.
Travel to T2 station	45 minutes	C1 reaches the next station at 11:45 AM.

Time component	Illustrative value	What it means
Rest requirement	30 minutes	C1 becomes ready at 12:15 PM.
T2 departure	1:00 PM	45 minutes remain as buffer.

A useful timing check
Ready time = previous duty finish + travel/transfer time + required rest
Basic feasibility: ready time <= required reporting or departure time
Buffer = departure time - ready time
In real life also add sign-on, preparation, safety and operational rules.

What if T1 is late? A quick what-if

Using the example above (ready time 12:15 PM, T2 departs 1:00 PM), see how a late T1 eats the buffer:

T1 delay	C1 ready at	Buffer left	Verdict
0 min	12:15 PM	45 min	Comfortable
15 min	12:30 PM	30 min	Comfortable
30 min	12:45 PM	15 min	Tight, check sign-on rules
45 min	13:00 PM	0 min	Zero buffer, very risky
60 min	13:15 PM	-15 min	Not feasible

Notice how quickly a safe plan becomes a risky one. A recovery system must keep re-checking times as delays arrive.

5 Crew availability and crew location

Two different questions that are easy to mix up

COACH There are **two** separate questions. (1) Is the crew **available** to work? (2) Is the crew already at the right station, or can it **get there in time**? A crew can say yes to the first and no to the second.

Example: train T2 departs Station B at 10:00 AM

Crew	Available from	Current location	Travel to B	Result	Feasible?
C1	9:00 AM	Station A	90 min	Arrives 10:30 AM, too late	**No**
C2	9:00 AM	Station B	0 min	Already at B	**Yes**
C3	10:30 AM	Station C	Unknown	Available only after departure	**No**

COACH Your turn. T5 departs Station C at **2:00 PM**. C1 is available at 1:00 PM at A and needs 90 minutes to reach C. C2 is available at noon at B and needs 45 minutes to reach C. C3 is available at 2:30 PM at C. **Which crews are feasible?**

YOUR ANSWER	Only C2. C1 is not feasible because he is available at 1:00 PM and needs 90 minutes to reach C, so he arrives at 2:30 PM. C3 is not feasible because he becomes available at 2:30 PM.
SAY IT LIKE THIS	Only C2 is feasible. C1 is not feasible because C1 becomes available at 1:00 PM and needs 90 minutes to reach Station C, so C1 arrives at 2:30 PM. C3 is not feasible because C3 becomes available only at 2:30 PM, after T5 departs.
COACH SAYS	Correct. C2 can reach Station C at 12:45 PM, well before the 2:00 PM departure.

COACH	Now the system question: **how could software find feasible crews automatically?**
YOUR ANSWER	The new crew should be available at the required station before the scheduled time, or after the scheduled time if the train is delayed. We should check which crew is eligible, available and can reach the station earlier.
SAY IT LIKE THIS	The recovery system should identify crews who are qualified and available, check their current locations, calculate their travel times, and confirm that they can reach the required station before the train's departure or reporting time. If the train is delayed, the system can use the revised time.
COACH SAYS	That is the right logic. Do not select a crew only because it is nearby. Qualification, availability, travel, rest and duty restrictions all matter.

The feasibility checklist

Check	The question to ask	Why it matters in mining
Qualification	Is the crew authorized for this train, route or locomotive?	Heavy-haul and steep routes often need special approval
Availability	Has the crew finished earlier work and become available?	Long cycles mean crews finish at the far end
Location	Where is the crew now?	Remote stations, few ways to move people
Travel / transfer	Can the crew reach the station in time?	Road, rail or air transfers can take hours
Rest and duty	Are rest and duty limits satisfied?	Long runs use up duty hours fast
Reporting buffer	Is there time for sign-on and preparation?	A late arrival may remove the buffer completely

6 Feasibility matrix and crew assignment

Putting the checks into a simple table of 1s and 0s

COACH Once we know which crew can drive which train, we can write it as a **feasibility matrix**. A **1** means the crew can operate the train. A **0** means it cannot. For now, each crew can drive **at most one** train in the planning period.

Crew	T1	T2	T3
C1	1	0	1
C2	1	1	0
C3	0	1	1

COACH How would you assign the three crews?

YOUR ANSWER Like C1 - t1, C2 - t2, C3 - t3.

SAY IT LIKE THIS I would assign C1 to T1, C2 to T2, and C3 to T3.

COACH SAYS Correct. All three pairings are feasible and no crew is used twice.

COACH Now suppose C2 suddenly becomes unavailable. What recovery assignment would you make?

YOUR ANSWER Then C1 - t1, C3 - t2 and C1 - t3.

SAY IT LIKE THIS I would assign C1 to T1 and C3 to T2. However, I cannot assign C1 to T3 as well because C1 is already assigned to T1.

COACH SAYS Good catch on the polished version. Under the one-train-per-crew assumption, C1 cannot cover two trains. With C2 out, T1 goes to C1 and T2 to C3, and **T3 has no remaining feasible crew**.

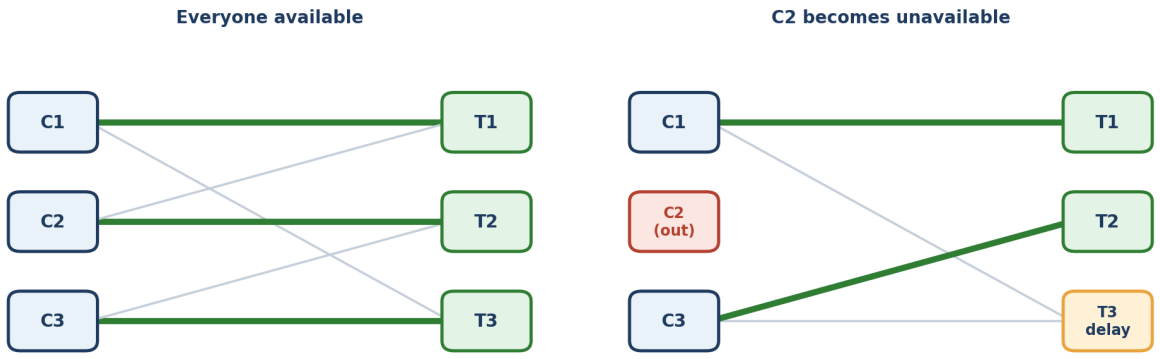


Figure 4. Left: the original plan. Right: after C2 drops out, T3 is left without a crew. Green lines show the chosen assignment, grey lines show other feasible pairs.

COACH	If T1 and T2 are high-priority trains but T3 is low-priority, which train would you delay, and why?
YOUR ANSWER	Then it's T3.
SAY IT LIKE THIS	Then I would delay T3 because it has lower priority.
COACH SAYS	Exactly. This is **priority-based recovery**: protect higher-priority trains and delay a lower-priority train when resources are not enough.

Why not delay a different train?
You could delay T1 or T2 instead, but then a high-priority train suffers for the sake of a low-priority one. When the model has no choice but to leave one train uncovered, it should be the one that costs least. Chapter 7 shows how the model makes this choice by itself with numbers.

7 Turning it into a MILP

A small optimization model, built piece by piece

COACH

MILP stands for **Mixed-Integer Linear Programming**. Don't let the name scare you. It simply means: describe your choices with variables, write your rules as constraints, say what you want to minimize, and let a solver find the best answer. We will build it in four short steps.

Step 1: Decision variables (the choices)

Variable	Meaning
$x_{ij} = 1$	Crew i **is** assigned to train j.
$x_{ij} = 0$	Crew i is **not** assigned to train j.
$D_j = 1$	Train j **is** delayed.
$D_j = 0$	Train j is **not** delayed.

Step 2: Objective (what we want to minimize)

Using illustrative delay penalties: T1 = ₹10,000, T2 = ₹8,000, T3 = ₹2,000.

Minimize $Z = 10{,}000\ D_1 + 8{,}000\ D_2 + 2{,}000\ D_3$
The model prefers to delay the train with the lowest penalty, as long as every other rule is satisfied.

Step 3: Constraints (the rules)

Rule	Math	Plain English
Each train is covered or delayed	$\Sigma_i\ x_{ij} + D_j = 1$ for every train j	Each train gets exactly one crew **or** is marked delayed. Not both.
A crew drives at most one train	$\Sigma_j\ x_{ij} <= 1$ for every crew i	One crew cannot be on two trains in this simple period.

Rule	Math	Plain English
Banned pairings stay banned	$x_{ij} = 0$ for infeasible pairs	A crew cannot take a train it cannot operate (the 0s in the matrix, plus unavailable crews).
Yes / no decisions	x_{ij}, D_j in {0,1}	Every decision is either yes (1) or no (0).

Step 4: Let the solver choose

COACH	Why do we need D_j? What does $\Sigma_i x_{ij} + D_j = 1$ mean? If the delay penalty for T1 is ₹2,000 and T3 is ₹10,000, which train would the model prefer to delay?
YOUR ANSWER	We need Dj to represent delay and help use to make object and constraints. This means either train has to assign with some crew or it is delayed. In that case T1 will go for delay.
SAY IT LIKE THIS	We need D_j to represent whether a train is delayed. It is used in the objective function and constraints. This means that each train must either be assigned to exactly one crew or be marked as delayed. In that case, T1 would be delayed because its delay penalty is lower.
COACH SAYS	All three answers are correct. The constraint makes sure a train is either given a crew or delayed, never both. The objective prefers the lower penalty, as long as other constraints stay the same.

Worked example: C2 is out, which option is cheapest?

Only C1 (T1 or T3) and C3 (T2 or T3) can work. The solver effectively compares options like these:

Option	Assignment	Delayed trains	Total penalty (Z)
A	C1 to T1, C3 to T2	T3	₹2,000 **(best)**
B	C1 to T1, C3 to T3	T2	₹8,000
C	C1 to T3, C3 to T2	T1	₹10,000
D	C1 to T1 only	T2 and T3	₹10,000
E	Nobody assigned	T1, T2, T3	₹20,000

Option A wins because it delays only the low-penalty train. A solver finds this automatically, even when there are hundreds of crews and trains.

Penalty comparison

Train	Delay penalty	Model's preference
T1	₹10,000	Avoid delaying if possible
T2	₹8,000	Avoid delaying if possible
T3	₹2,000	More acceptable to delay in this example

Important modelling note

These penalties are teaching numbers, not real money and not a recommended business objective. A realistic model may combine delay minutes, train priorities, overtime, reassignment costs, deadheading cost and other measures.

Try it in Python (optional)

This is the same model written with the PuLP library. I tested it with PuLP 2.9 and it returns option A with Z = 2,000. If a newer PuLP version raises errors on this syntax, install it with: pip install pulp==2.9.0

```
import pulp
crews  = ["C1", "C2", "C3"]
trains = ["T1", "T2", "T3"]
ok = {("C1","T1"):1, ("C1","T2"):0, ("C1","T3"):1,
      ("C2","T1"):1, ("C2","T2"):1, ("C2","T3"):0,
      ("C3","T1"):0, ("C3","T2"):1, ("C3","T3"):1}
pen = {"T1": 10000, "T2": 8000, "T3": 2000}
available = {"C1": 1, "C2": 0, "C3": 1}      # C2 is out

m = pulp.LpProblem("crew_recovery", pulp.LpMinimize)
x = pulp.LpVariable.dicts("x", (crews, trains), cat="Binary")
D = pulp.LpVariable.dicts("D", trains, cat="Binary")

m += pulp.lpSum(pen[j] * D[j] for j in trains)           # objective
for j in trains:                                          # covered or delayed
    m += pulp.lpSum(x[i][j] for i in crews) + D[j] == 1
for i in crews:                                           # at most one train
    m += pulp.lpSum(x[i][j] for j in trains) <= 1
for i in crews:                                           # banned pairs
    for j in trains:
        if ok[(i, j)] == 0 or available[i] == 0:
            m += x[i][j] == 0

m.solve(pulp.PULP_CBC_CMD(msg=0))
print(pulp.LpStatus[m.status], pulp.value(m.objective))
```

8 When a late train hits the next assignment

The ripple effect that makes recovery hard

COACH Here is the real trap. T1 departs Station A at 8:00 AM and should arrive at Station B at **11:00 AM**. T2 departs from Station B at **12:30 PM**. C1 drives T1 and needs **30 minutes** of rest after arriving. C2 is already at Station B, available from 7:00 AM.

COACH If T1 arrives on time, C1 is ready at 11:30 AM. That is a one-hour buffer before T2. Now T1 arrives **late at 11:45 AM**. After the 30-minute rest, C1 is ready at 12:15 PM. Only **15 minutes** remain, and we must check whether that is enough under the sign-on and operating rules.

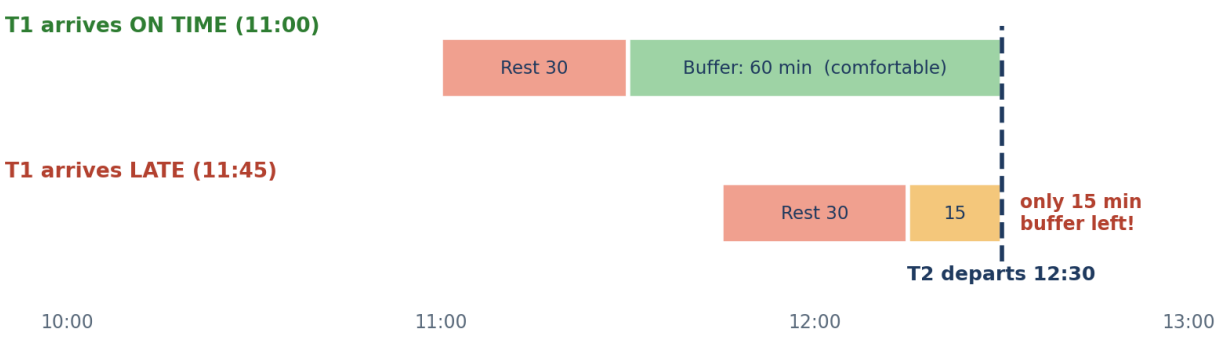


Figure 5. One late train turns a comfortable 60-minute buffer into a risky 15-minute one.

Field reality: this is the cascade
In a mining cycle the same crew, locomotive and track slot are often chained from one run to the next. One late arrival at the port does not hurt only that train. It squeezes the next crew assignment, the next train's departure and sometimes a ship window. That is why recovery tools look ahead and not only at the current train.

Your next exercise

COACH

T1 was due at 11:00 AM but arrives at **11:45 AM**. C1 drove T1, needs 30 minutes of rest, and is already at Station B. T2 departs B at **12:30 PM**. C2 is available at B from **11:00 AM**.

Q1. At what time is C1 available after T1?
Q2. Can C1 operate T2 at 12:30 PM, if C1 must be ready at departure?
Q3. Can C2 operate T2? Which crew would you prefer, and why?

Pause here and write your own answers first.

Check your thinking

COACH SAYS

Q1. 11:45 + 30 min rest = **12:15 PM**.
Q2. Yes on paper: 12:15 is before 12:30, leaving only a **15-minute buffer**. Whether that is truly enough depends on sign-on and preparation rules. If sign-on needs 20 minutes, the answer becomes no.
Q3. Yes, C2 is at B and free from 11:00 AM, so C2 has a long buffer. C2 looks safer. But the best choice also depends on what C2 and C1 are planned to do next, qualifications, duty hours, and the objective. Do not choose from this information alone.

This is a model answer to learn from, not a rule. The right choice always depends on the full operator rules and on the objective you set.

9 From toy model to real model

What changes when we leave the classroom

COACH

Our first MILP is deliberately tiny. It assumes **one crew, one train**. A real crew can drive several trains in a row. Let's see what real systems add.

Toy model	Real model adds	Why
One train per crew	A **sequence** of trains per crew	Crews work shifts with several runs
Times checked by hand	**Time-expanded network**: every station and time slot as a node	Lets the model reason about when and where
Crew just has a location	Depots, relief points, crew change points	Crews start, end and swap at specific places
Travel is a number	**Deadheading** as a real option with cost	Moving crews costs time and money
Simple rest	Legal rest and duty rules, maximum hours, fatigue limits	Safety and law
All crews can drive all trains	Qualification by route, locomotive, train type	Real competency limits

Toy model	Real model adds	Why
Delay is yes/no	**Delay minutes** and propagation to later trains	A 5-min delay is not a 5-hour delay
Only crew cost	Reassignment cost, overtime, cancellation, ship-window penalties	Closer to real business goals
One decision moment	Re-run as new disruptions arrive; keep plan stable	Recovery is continuous

What a good recovery model tries to balance

- **Delay** (weighted by train priority and ship windows)
- **Plan stability**: change as few crews as possible, because every change disturbs people
- **Cost**: overtime, deadheading, standby crews
- **Speed**: control rooms need an answer in minutes, not hours
- **Always safe**: never break hard rules, even to save money

Data is half the battle

A perfect model fails with bad data. Real work often means getting correct crew positions, live train times, qualification lists and rule definitions. Confirm all definitions and rules in the operator's own documents before building anything real.

10 Key terms, quick answers and roadmap

Your revision sheet

Key terminology

Term	Simple meaning
Crew scheduling	Planning crew assignments before operations take place.
Crew recovery	Changing the plan after a disruption.
Disruption	An event that affects the original plan, such as crew absence or a train delay.
Crew availability	The earliest time a crew can start another assignment.
Crew location	The station or place where the crew currently is.
Travel / transfer time	Time needed for the crew to reach the next train's departure station.
Rest period	Required rest between duties under the applicable rules.
Duty time	Work time counted under the operator's rules.
Qualification	Required competence or authorization for a train, route or locomotive.
Feasibility	Whether an assignment satisfies all required conditions.
Buffer time	Extra time between the crew becoming ready and the next required reporting or departure time.
Deadheading	Moving a crew to another location without that crew operating the train during that movement.

Term	Simple meaning
Reassignment	Changing a crew's original train assignment.
Decision variable	A variable representing a choice the optimization model makes.
Binary variable	A variable that can only be 0 or 1.
Objective function	The expression the model tries to minimize or maximize.
Constraint	A rule that every valid solution must satisfy.
Feasible solution	A solution that satisfies all constraints.
Optimal solution	A feasible solution with the best objective value.
MILP	Mixed-Integer Linear Programming: linear objective and constraints with some integer or binary variables.
Hard constraint	A rule that can never be broken (safety, law).
Soft goal	A preference the model tries to satisfy as well as it can.

Short technical answers (say these out loud)

Question	Suggested answer
What is the first thing to check before assigning a crew?	Check whether the crew is feasible: available, qualified, able to reach the departure station, and compliant with duty and rest rules.
Why is a crew's location important?	A crew may be available but too far away to reach the required station before the train departs.
What is buffer time?	The extra time between the crew becoming ready and the required reporting or departure time.
Why can a train delay affect another train?	The same crew may be planned to operate a later train. A late arrival can reduce or remove the time available for the next assignment.
What does x_{ij} represent?	It is 1 if crew i is assigned to train j and 0 otherwise.
What does D_j represent?	It is 1 if train j is delayed and 0 otherwise.
What does $\Sigma_i x_{ij} + D_j = 1$ mean?	Each train must either receive exactly one crew or be marked delayed.
Why do we need an objective function?	It tells the optimizer what outcome to prefer, such as minimizing delay penalties or operating costs.
What if no feasible crew is available?	The model may need to delay or cancel the train, or use another permitted recovery option if one exists.
Why should we not assign the nearest crew automatically?	The nearest crew may be unavailable, unqualified, or unable to satisfy duty and rest requirements.

Recommended learning roadmap

Step	Next topic	Why it matters
1	Complete the late-arrival exercise	Practise readiness time, buffer and choosing between crews
2	Sign-on, sign-off, duty duration and rest rules	Turn simple timing checks into realistic rule checks

Step	Next topic	Why it matters
3	Crew depots, relief points and deadheading	Show where crews start, end and move
4	Sequential assignments with time and location constraints	Let one crew drive several trains in a valid order
5	Delay minutes, reassignment costs and train priorities	Build a more realistic recovery objective
6	Implement a small model in Python with PuLP or Pyomo	Turn the formulation into a working model

Modelling boundaries to remember

The first MILP assumes each crew drives at most one train. Real recovery needs sequences of duties, time-expanded assignments, location changes, legal rest and duty rules, qualifications, relief points and detailed delay propagation. The examples here do not establish actual operator rules. Confirm data definitions, constraints, priorities and permitted recovery actions in the relevant company or railway documents before implementing anything real.

END OF GUIDE